

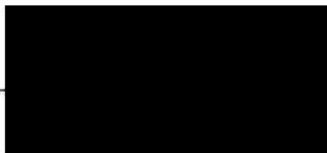
ChE422 Biochemical Engineering

2020 Midterm Exam

3/10/2020

2:30 PM - 3:45 PM

Name: _____



75%
10

Instructions: This is a close-book and close-note exam.

MULTIPLE CHOICE QUESTIONS (Part I):

Note: Only ONE answer is correct. Circle the correct answer. (2 pts for each problem)

1. The bond between a phosphate group and the ribose sugar group in RNA is called which of the following?
☒ (A) Phosphodiester linkage
(B) Glycosidic bond
(C) Peptide bond
(D) Amide bond
2. Which of the following plays a role in DNA transcription?
☒ (A) DNA polymerase
(B) RNA polymerase
(C) DNAase I
(D) Guanyl transferase
3. What signals the end of transcription?
☒ (A) Stop codon
☒ (B) Terminator
(C) The end of the DNA chain
(D) RNA polymerase runs out
4. The start codon is:
(A) AUU (B) UAA ☒ (C) AUG ☒ (D) UAG
5. Which of the following is not a component of a nucleotide?
☒ (A) Phosphate group
(B) Anti-codon
(C) Ribose sugar
☒ (D) Nitrogen base
6. What provides the energy that drives the addition of nucleotides to a growing DNA chain during replication?
(A) The hydrolysis of peptides
☒ (B) The hydrolysis of the diphosphate group in dNTPs
(C) The hydrolysis of UTP
(D) The hydrolysis of GTP

7. Which of the following is the sequence of the -10 region of prokaryotic promoters?

- (A) TATAAT (B) TTGACA (C) TATA (D) CAA

8. How many base pairs make up a codon?

- (A) 1 (B) 2 (C) 3 (D) 4

9. How many amino acids are there?

- (A) 20 (B) 30 (C) 40 (D) 50

10. How many hydrogen bonds are formed between one A:T base pair?

- (A) 1 (B) 2 (C) 3 (D) 4

MULTIPLE SELECTION QUESTIONS (Part II):

Note: ONE or MORE answers can be correct. Circle all correct answers. (3 pt for each problem)

11. Which of the following are the protein secondary structures?

- (A) Hairpin (B) α -helix (C) β -Sheet (D) Peptide bond

12. Which of the following in a plasmid controls the copy number of this plasmid in the host?

- (A) Antibiotic resistance gene
(B) Multiple cloning sites
(C) LacZ gene
(D) Origin of replication (Ori)

13. Which of the followings are required for Polymerase chain reaction (PCR)?

- (A) DNA template (B) T4 ligase (C) DNA polymerase (Taq) (D) DNA primers

14. Which of the following methods are commonly used to transform plasmid into *E. coli* cells?

- (A) Cell fusion
(B) Heat shock method
(C) Electroporation
(D) DNA injection by a needle

15. Which of the followings are the key components in a general Operon?

- (A) Promoter (B) Operator (C) phage gene (D) Structural genes

16. Which of the following methods are used in random mutagenesis?

- (A) Error prone PCR
- (B) Chemical Deamination
- (C) DNA shuffling
- (D) *E. coli* XL1 Red method

17. Agarose gel electrophoresis is able to separate linear DNA fragments with which of the following parameter?

- (A) Charge
- (B) Size
- (C) Secondary structure
- (D) Conformation

18. In order to use pET expression system for protein expression, the host cell should provide with which of the following component for transcription?

- (A) Taq DNA polymerase
- (B) T7 polymerase
- (C) protease
- (D) Histag

19. Which of the following methods can be used to introduce the mutation at the specific sites in a gene?

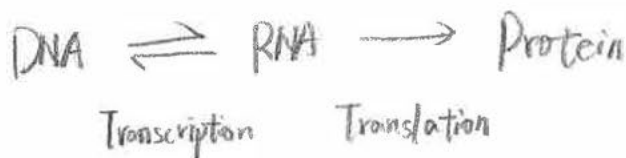
- (A) UV irradiation
- (B) Cassette mutagenesis
- (C) PCR site-directed mutagenesis
- (D) RACHITT method

20. Which of the following compound is used in pBAD expression system to induce protein expression?

- (A) Glucose
- (B) IPTG
- (C) L-arabinose
- (D) Lactose

Brief answer the following questions

1. What is Central Dogma of Molecular Biology? (5pts)



2. List three key differences between prokaryote and eukaryote? (5pts)

① the eukaryote has the nuclear core that 有核膜包被的细胞核 covered by membrane, while prokaryote doesn't have it.

② the eukaryote has larger size than prokaryote.

③ the mRNA must be transferred through the nuclear membrane if it wants to translation, while

3. Describe the general key components of an expression plasmid? (5pts) prokaryote doesn't need to do so.

Antibiotic resistance gene

Multiple cloning sites, a plasmid have at most 20 restriction sites

LacZ gene

Origin of replication

4. Describe the key differences between blunt-end ligation and sticky end ligation for cloning? (5pts)

Sticky-end ligation can be controlled, while blunt-end ligation can NOT be controlled

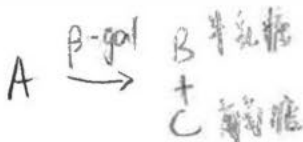


5. List at least two methods to confirm the insertion of a gene into the expression plasmid? (5pts)

① DNA sequencing.

② blue-white screening. If inserted, the intact structure is broken and make result become white. If not inserted, then become blue.

6. Briefly describe blue-white screen? (10pts)



Usually, β -gal has specific gene that have antibiotic resistance and can be alive in the cell. After left reaction, the gene is intact and make the result become blue. If the plasmid is inserted and the structure is not intact, it will not generate final product and make the result white.

7. After you find an interesting gene in a new organism, briefly describe the procedures to let *E. coli* express this protein. (15pts)

- ① use target gene to make DNA Template and generate a large amount of DNA by using PCR Method.
- ② split the DNA into two strands with DNA polymerase, and then insert these target gene into plasmid.
- ③ inserted plasmid is transferred into *E. coli* cell and break the circle into a line.
- ④ These line works as tRNA, and in ribosome, they will be encoded into amino acid and protein.

